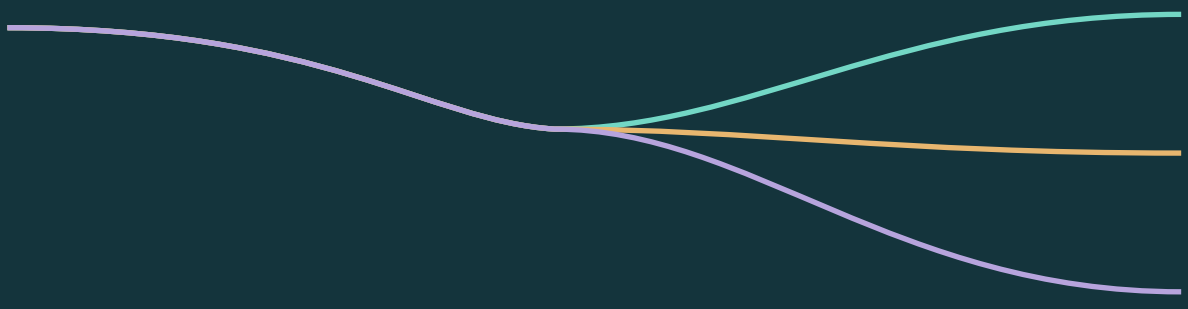
RENALIS / RESEARCH DOSSIER

Evidence. Decisions. The next twin.

A detailed strategy for a dialysis digital twin that exposes competing explanations and asks which observation could clarify the comparison.



HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

Research synthesis and proposed master plan | 17 October 2026

Product implementation remains unchanged. No clinical efficacy is established.

01 / DIRECTION

The decision and its rationale

Keep the dialysis problem. Deepen the representation, evidence and clinical workflow.

Recommended direction: a dialysis twin that preserves competing physiological explanations, shows when they imply different futures, and tests which feasible observation could clarify the comparison.

RENALIS already has a working research sandbox, explicit model disagreement, observation-value calculations, safeguards and saved evidence. The next step is to make those behaviors scientifically stronger and much easier to understand. A whole-body pivot would move away from the organizer's localized-outcome requirement without providing a better verified model foundation. [R01] [L04]

Five decisions that organize the master plan

Decision	Reason and proof needed
Retain maintenance-dialysis scope	One chronic condition, a defined future event, recognizable clinical context and relevant available data.
Learn from preceding sessions	Current authored priors do not establish learned patient memory. Demonstrate incremental value with fair temporal ablations.
Make physiology inspectable	Use a connected computational graph and competing models. A graph or close curve fit alone does not identify a cause.
Evaluate practical observation policies	Include availability, delay, failure and burden. Better information and better decisions are separate outcomes.
Unify evidence and communication	One patient, one cutoff, readable explanations and a reproducible submission bundle. Keep observed, simulated and proposed results distinct.

Ambition is not capped by builder count. The roadmap extends from a strong hackathon release to multi-session learning, paired physiological data, multi-centre validation and an appropriately governed impact study. Progress is gated by evidence and dependencies, rather than an arbitrary limit on features.

This dossier is a research and product plan. It neither guarantees a competition result nor establishes clinical safety, treatment effects or patient benefit. No product implementation was changed to create it.

01 / DIRECTION

Reading map and report scope

A navigable synthesis of the recent requests, with explicit evidence labels.

01 / Direction
Pages 2-2
Recommendation, scope and evidence labels.

02 / Competition
Pages 4-6
Requirements, judging inference and mandatory artifacts.

03 / Comparative research
Pages 7-11
Winners, finalists, clinical prior art and originality.

04 / Current evidence
Pages 12-15
Implemented engines, retained results and failure hypotheses.

05 / Reference repositories
Pages 16-18
Seven references, adoption choices and licensing.

06 / Target design
Pages 19-27
Architecture, graph, memory, counterevidence and workflow.

07 / Validation scenarios
Pages 28-29
Realistic patient questions and operational constraints.

08 / Evaluation protocol
Pages 30-33
Factual, mechanism, policy and clinical-evidence studies.

09 / Master plan
Pages 34-38
Dependency roadmap and 26 detailed work packages.

10 / Explanation and submission
Pages 39-40
Full demo storyboard and clear narrated explanation.

11 / Decisions and next actions
Pages 41-43
Decision records, review triggers, risks and glossary.

12 / Reference register
Pages 44-50
Linked primary sources and local evidence register.

The four most recent requests before this report covered: reference repositories and competitors; deep research on differentiation and judging; reusable pitch-video guidance; and voice, motion and screen-clarity improvements. The main body consolidates the research requests. The communication requests inform the demo and submission chapters. Earlier originality and UI decisions are retained as context. [L06]

Evidence labels: VERIFIED = published or directly inspected source; IMPLEMENTED = present in the inspected project; RESULT = retained experiment output; PROPOSED = planned work; OPEN = unresolved question. A recommendation is an inference from evidence, not an organizer instruction.

Method: focused primary-source web review, read-only repository inspection and review of retained RENALIS reports. External code was not executed. Competitor results were not independently reproduced. Source review is not an exhaustive systematic, patent or commercial-product search. Click source IDs to reach the reference register.

02 / COMPETITION

Work backward from the brief

The mandatory task is a localized predictive twin with meaningful two-stream fusion.

Requirement	RENALIS interpretation	Evidence to show
One chronic/lifestyle condition	CKD requiring maintenance hemodialysis	Specific patient population and Indian relevance.
Localized future adverse event	A new measured low-SBP event during a session	Exact threshold, horizon, eligibility and future reveal.
Historical/static stream	EHR plus completed preceding sessions	Actual influence on state or prediction, with ablation.
Dynamic stream	Available BP and dialysis-machine observations	Timestamp-consistent replay and sensor provenance.
Working algorithm and doctor UI	Updating state, forecast and evidence workspace	Live inputs change outputs; assumptions are inspectable.
Allowed sandbox data	Synthetic or eligible anonymized/open data	Provenance, license, data dictionary and exclusions.

The public listing places a whole-body twin outside scope. A physiological graph can still be rich: it should deepen the selected dialysis question rather than add unrelated diagnoses. Dedicated medical IoT is allowed, so a dialysis-device stream is a defensible fit to the brief. [R01] [L01]

Verified calendar, checked 17 October 2026

Registration: 18 October, 7:00 PM IST. Prototype/code: 20 October, 7:00 PM IST. Virtual jury date: 30 October. Summit showcase: 20 November in Bengaluru. These external deadlines should guide release readiness; they do not cap the long-term scientific roadmap. [R01]

OPEN: the supplied form says 15-20 minutes while the brief says minimum 20 minutes. Plan a 20:00 final demo and clarify the boundary before export. Shortlisting language also differs across paragraphs; do not invent a resolution.

02 / COMPETITION

What we can infer about judging

A proposed evidence matrix, not an invented official scorecard.

No numeric rubric or previous winners of this exact Happiest Health challenge were verified. The following review questions are inferred from its published requirements. Another competition's scores are useful context but are not this organizer's rules. [R01]

Review question	Evidence the judge should be able to inspect
Does it meet the task?	One condition; one exact event; two data streams; updating algorithm; clinician dashboard.
Is the fusion substantive?	A feature/provenance map and controlled ablation showing the effect of preceding information.
Is the prediction prospective in time?	A cutoff marker, unavailable future, input manifest and later factual reveal.
What is distinctive?	Close historical fits, diverging scenario responses, predicted measurement ranges and a real update.
Does it work under constraints?	Unavailable sensors, delayed or failed acquisitions, stable cases and unresolved states.
Are the results credible?	Fair baselines, held-out units, uncertainty, complete failures and transparent source claims.
Can we understand and reproduce it?	Readable patient story, one-command synthetic run, versioned results and accessible artifacts.

Evidence beats feature counting

A new module earns its place when it answers one of these questions and has a measurable completion condition. The physiological graph must reflect actual computation; historical memory must be estimated from preceding data; the observation selector must face a fair no-observation comparison. These are proposed project standards, not asserted judging weights.

The final presentation should place the relevant evidence next to the claim. A source citation supports a background rationale; a RENALIS experiment supports a RENALIS performance claim. Neither substitutes for the other.

02 / COMPETITION

The mandatory submission ledger

Treat completeness as a separate deliverable with explicit verification.

README item from the form	Current position and planned completion
Team details	Final verified details required; include only authorized public information.
College / incubator	Final affiliation details required.
Project title	RENALIS exists; sharpen the descriptive subtitle.
Problem statement	Present; align exact clinical problem and intended user.
Healthcare use case	Present; add one coherent end-to-end scenario.
Technical stack	Present; match the final release.
AI/ML framework details	Present; explicitly distinguish learned, authored and simulated components.
15-20 min unlisted YouTube link	Current 2:46 film is insufficient. Plan 20:00 and verify duration/access.
Open-source license	Local license present; add notices for any adopted external code.
Architecture PDF / PPT	Required-format artifact still needed; this dossier is not the dedicated architecture submission.
Project presentation PDF / PPT	A concise judged presentation is still needed; the research dossier is supporting material.
Public files and links	Signed-out accessibility remains to be verified before submission.

The brief also requests a Team Name_College Name folder and a public repository. Link every result, deck and recording to the final code/model revision. README presence alone does not verify public playback, downloadable files or reproducibility. [LO1]

This report does not publish the repository, upload a video, register a team, submit an entry or resolve missing identity details. Those remain separately authorized external actions.

03 / COMPARATIVE RESEARCH

What verified winners make clear

Use documented examples to study execution, without claiming to know jury deliberations.

Example and verified result	Documented problem	Interpretation for RENALIS
HospiTwin / MIT 2025	InterSystems GenAI challenge award; emergency-department boarding and bottlenecks.	Explain a recognizable workflow and its users. This was a named challenge award, not verified overall first place. [R20]
CareCast / NYC 2025 first	Personalized climate-health alerts and access to resources.	Connect prediction to a complete user journey. [R21]
Verdant / NYC 2025 second	Healthcare-product emissions predictions and lifecycle contributions.	Make model outputs answer a concrete decision question. [R21]
HeatIQ / NYC 2025 third	Localized heat forecasts connected to outdoor work-site planning.	Define who acts, where and over what time horizon. [R21]

The NYC 2025 organizer publishes impact, team, AI/data, innovation, feasibility and scalability dimensions. That supports a broader lesson: novelty is one part of an entry's execution. It does not establish Happiest Health's weighting or predict which project will win. [R21]

What this changes in our presentation

Open with a patient and a specific uncertainty during a dialysis session. Introduce the system when it answers that uncertainty. Reveal technical depth through a functioning workflow, rather than beginning with a list of frameworks. Give the audience a visible outcome and a clearly stated remaining limitation.

The inferred success pattern is coherence: a defined user, an important problem, a purposeful model, an observable workflow and evidence suited to the claim. It is not a causal explanation of past awards.

03 / COMPARATIVE RESEARCH

Winners versus other finalists

Compare public descriptions fairly; finalist status is not evidence of a poor project.

NYC 2026 organizer result	Publicly described focus	Useful design signal
Chamomile / first	Sensors, buildings and occupants connected through a data platform and business model.	An integrated hardware/software workflow and route to adoption.
Grounded Cyborg / second	Individual health and prescription data within existing clinician workflows.	Explicit workflow fit.
HeatSafe / third	Heat-risk forecasting for Medicare/Medicaid populations with pharma/EHR context.	Defined population and deployment context.
Climate Compass / finalist	Heat and air-quality information for daily planning.	Clear environmental context; deeper workflow detail is not established by the short description.
Five Degrees / finalist	Green-space access as preventive care.	A recognizable preventive-care use case.
CareBuilders / finalist	Canopy/air-quality-informed routes.	A tangible planning interaction.

These are another organizer's 2026 results. They provide a same-event comparison between awarded teams and other listed finalists. Public pages do not reveal complete demonstrations, technical evaluations or scores; they cannot prove that workflow specificity caused the placements. [R22]

A practical response for RENALIS

Specify the immediate user as a clinician or researcher reviewing evidence sufficiency during maintenance dialysis. Show the handover, timestamps, available equipment and the next review step. A research sandbox can demonstrate workflow fit while remaining explicit that clinical recommendations and outcome improvement have not been established.

Do not label the user-supplied GitHub projects as winners, non-winners or confirmed entrants. Their public repository names and features establish none of those statuses.

03 / COMPARATIVE RESEARCH

The prior-art boundary

Forecasting, digital-twin language, adaptive removal and informative sampling already have precedent.

Existing work	What it changes in our claims
Real-time IDH prediction	EHR plus machine-stream forecasting is established; it remains essential task evidence. [R05]
IoMT / EMR digital-twin platform	Dialysis digital-twin terminology predates this project. [R06]
BestShape follow-up	A deployed prediction-and-response workflow has been evaluated retrospectively. Dashboard alerts alone are not sufficient differentiation. [R07]
bioLogic Fusion	The manufacturer describes adaptive UF using SBP and relative blood volume. Multi-sensor fluid-management adaptation is existing product territory. [R08]
Sampling-design studies	Choosing informative samples in dialysis-related models is established adjacent methodology. [R16]
SICATA survival twin	A conference abstract uses digital-twin terminology for survival modeling; evidence and task differ from ours. [R45]

BioTwin Omni publicly pitches multi-agent deliberation, hypothetical interventions, crisis simulation and a 3D dashboard, and mentions the challenge. That is self-reported positioning; execution, submission and awards were not independently established. Diaprobe describes adjacent sensing/stimulation research and an under-development product. Neither source proves clinical superiority. [R39] [R40]

Defensible contribution: an inspectable implementation of decision-relevant physiological uncertainty, feasible additional observation and retained unresolved states. Global priority and clinical benefit remain unverified.

A more exhaustive novelty assessment would expand to patents, commercial documentation, systematic database searches and expert interviews. Similar ideas can use different terminology; absence from this focused search is not proof of absence.

03 / COMPARATIVE RESEARCH

Why the physiological question matters

A pressure curve is evidence about a system; it is not a unique explanation of that system.

The hemodynamic study identified cardiac-power, peripheral-resistance and combined patterns during hypotensive episodes. Mechanisms can coexist. RENALIS's volume/refill, vascular and cardiac families are simplified authored models, not a validated classification of those clinical groups. [R03]

KDIGO emphasizes individualized BP/volume assessment and competing risks from instability and fluid excess. This supports displaying unfinished fluid removal alongside pressure scenarios. It does not justify an automatically selected safe profile or claim of prevented mortality. The 2025 UK dialysis BP guideline belongs in clinical expert review. [R02] [R44]

Research finding	Design consequence
Better curve fit can require implausible physiology	Constrain parameters, inspect identifiability and evaluate scenario errors separately. [R13]
Intermittent cuffs miss events seen by continuous monitoring	Use a measured-event endpoint; do not label missing intervals as observed stability. [R12]
Initial and recurrent IDH change task difficulty	Separate event types and compare against a prior-occurrence baseline. [R11]
Measurement performance depends on patient and method	Model channel quality and patient shifts; do not treat authored noise as a device specification. [R14] [R42]
More monitoring need not improve outcomes	Evaluate the downstream policy and later patient impact, not only information gain. [R10]

Indian relevance without invented burden statistics

The 23,601-patient Indian private-network study supports local care relevance and centre variation. It is not a national prevalence estimate or evidence that IDH explains survival differences. Resource scenarios should be grounded through local clinicians and site observations; practice-study variation motivates that investigation. [R09] [R41]

03 / COMPARATIVE RESEARCH

What is distinctive - and what is not

Originality should be demonstrated through behavior and tested against meaningful alternatives.

Component	Status of the idea	Our research opportunity
Predict low BP ahead	Established research	Precise endpoint, reproducible baselines and useful lead time.
Fuse EHR and dynamic signals	Established requirement / prior art	Genuine temporal history learning and measured incremental value.
Represent physiology as a graph	Established research approach	A computationally faithful, readable explanation of model assumptions.
Retain several mechanisms	Known modeling principle / physiology	Expose when compatible pasts imply different scenario comparisons.
Choose another observation	Established information/design methods	Decision value under real availability, delay, failure and burden.
Show unresolved or contradicted states	Known uncertainty / abstention concepts	Evaluate false certainty and useful abstention in a complete dialysis workflow.

The candidate contribution is the combination: shared factual evidence, rival responses, a decision-relevant observation question, a timestamp-correct update and a visible unresolved outcome when needed. We did not establish that no other system contains that combination. [R03] [R05] [R16] [R23]

The claim we can use today

In a synthetic workspace, RENALIS retains authored physiological models, exposes scenario disagreement and calculates expected observation value under declared assumptions. The current challenge does not show decision-value superiority. [L03] [L04]

The claim we aim to earn

On independently specified tasks, feasible observation selection improves a declared research decision objective while controlling false certainty and burden. A future clinical-benefit claim would need its own appropriately governed evidence; it cannot be earned by a more polished animation.

04 / CURRENT EVIDENCE

What is implemented now

There are separate engines and experiments. Their results must not be blended.

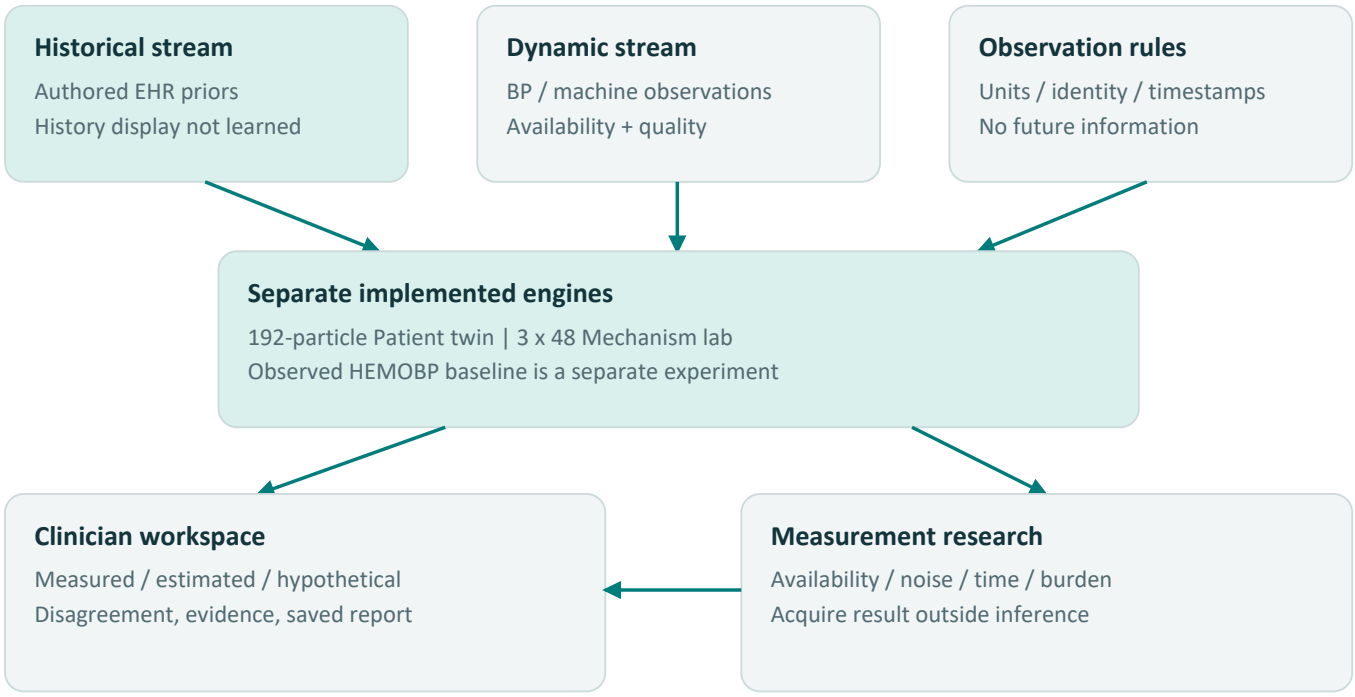


Figure 1. Current project organization, simplified from inspected implementation. The retrospective baseline is separate from the synthetic twin; it does not train or clinically validate the displayed hypothetical responses. [L04] [L05]

Implemented element	What it establishes and what it does not
Patient twin: 192 particles	Tracks a two-compartment fluid state using authored priors and available BP/machine history. Conditional model uncertainty is not calibrated clinical risk.
Mechanism lab: 3 x 48 particles	Fits authored response families and compares hypothetical profiles. Compatibility is within the chosen model set, not a diagnosis probability.
Safeguards and evidence reports	Handles stale data, current low BP, completion and contradiction; retains immutable saved evidence. Research policies are not validated clinical protocols.
Observed HEMOBP baseline	A separate retrospective predictive experiment with held-out patients. It does not establish effects of changing a removal profile.

Carry-over currently accounts for unfinished target plus user-authored net gain. It is not a validated longitudinal physiological model. The existing landing page, guide and short film are communication assets; their presence does not establish clinical usability. [L04]

04 / CURRENT EVIDENCE

Observed-data result: useful signal, gaps remain

Read the retained benchmark as a specific retrospective task, not a clinical warning-system validation.

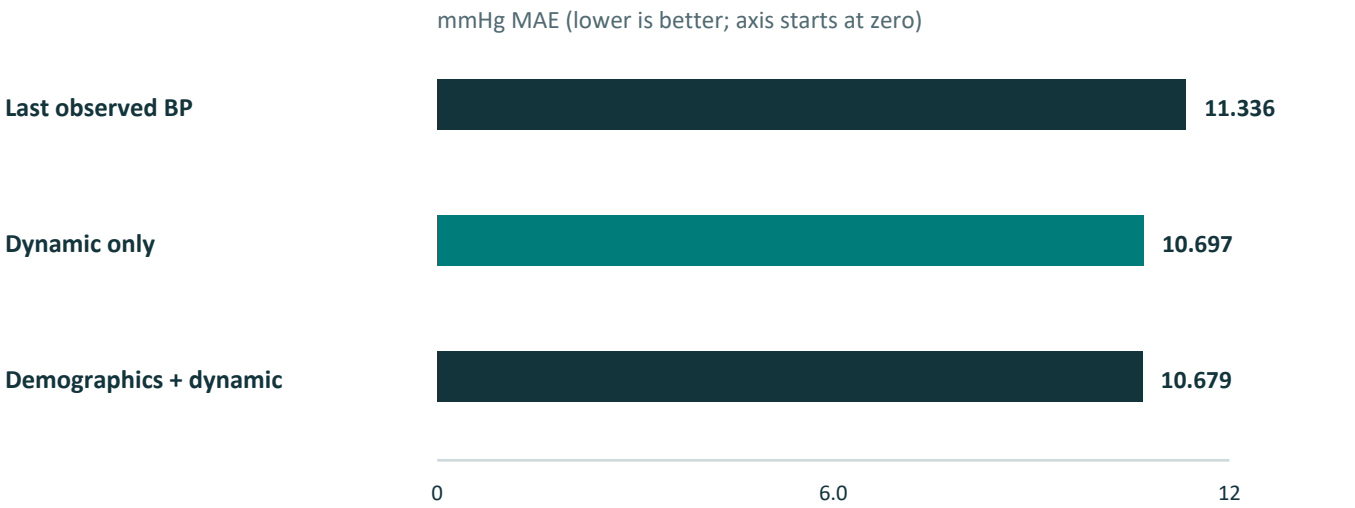


Figure 2. Retained RENALIS test MAE. All bars start at zero. Demographic addition improves MAE by 0.018 mmHg over dynamic-only; statistical or clinical importance has not been established. [L02]

Quantity	Retained value / interpretation
Test population	211 patients; 77,138 windows; no training/test patient overlap.
Task	Next distinct measured SBP, gap 5-35 min, current SBP >90; median gap 19.97 min.
Event results	Sensitivity 0.753; PPV 0.127; PR AUC 0.197; 11,497 false-positive windows.
Meaning of historical input	Observed fused baseline adds age, sex and diabetes; it does not yet learn rich preceding-session history.
Unresolved audit	Session inference, repeated-measure identity, UF units, informative sampling and alert consolidation.

The test includes 2,212 positive windows. Window-level false positives are not a count of clinical alarms. The endpoint is not exactly an anytime-within-30-minute event and does not isolate first events from recurrence. Raw event scores have not been assessed as calibrated clinical probabilities. [L02]

Next proof: audit the cohort, lock endpoints and splits, add genuine preceding history, then report paired uncertainty, calibration, lead time and consolidated workload. Keep this benchmark separate from scenario-effect claims.

04 / CURRENT EVIDENCE

The central policy result is not a win yet

The independent synthetic suite is valuable because it exposes a failure of the intended claim.

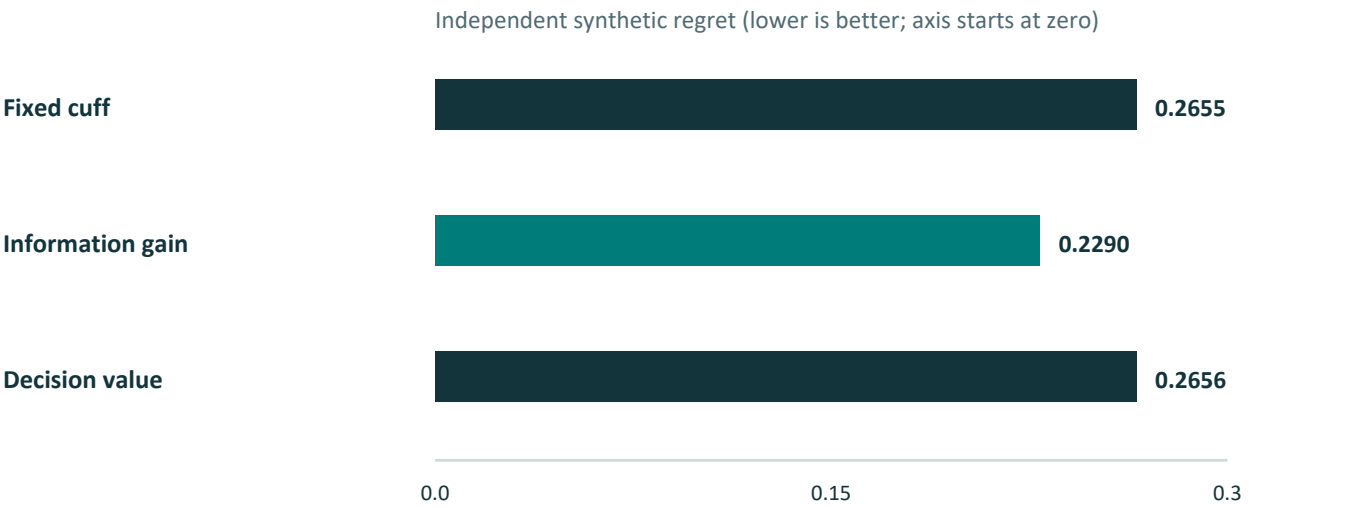


Figure 3. Retained independent synthetic regret, lower is better. Decision value is 0.2656 versus fixed cuff 0.2655; average superiority is not demonstrated. No uncertainty interval was retained with these summaries. [L03]

Policy	Mean regret	Acquisitions	Worse than cuff
Fixed cuff	0.2655	108 / 108	Reference
Family information gain	0.2290	108 / 108	8 cases
Decision value minus burden	0.2656	96 / 108	8 cases

The suite contains 108 cases: 36 shared-past histories crossed with three hidden response types. Scoring futures use independently authored additive-response equations rather than the twin's compartment-pressure equation. Policies receive the same pre-generated value if they choose the same channel and share an at-most-one immediate-observation budget. [L03]

This reduces one kind of circular self-scoring, but remains synthetic and shares the declared research loss. It does not test delayed acquisition or clinical outcomes. Information-only selection has lower retained mean regret here, while both adaptive approaches make eight cases worse. Family discrimination and better scenario choice must be evaluated separately.

Preserve this suite as a diagnostic result. Changes motivated by it are development work. A new predeclared holdout is needed before claiming improved generalization.

04 / CURRENT EVIDENCE

Explain failures before expanding claims

Candidate causes, discriminating tests and the consequence of each result.

Candidate explanation	Test that could distinguish it	Consequence
Model mismatch	Compare within-family and independently generated mixed/omitted-mechanism futures at matched budgets.	If performance collapses under mismatch, improve representation and abstention rather than sensor ranking alone.
The research loss drives the result	Sweep predeclared loss weights on development data; freeze separate evaluation settings.	Publish sensitivity and tradeoffs instead of selecting the one setting that wins.
Channel proxies are too convenient	Shift bias, correlation, noise, overlap and acquisition failure independently of the estimator.	Estimate how much apparent value depends on authored sensor equations.
Numerical integration changes ranking	Compare three-node quadrature with higher-order or seeded Monte Carlo references.	Refine computation when ranking errors are material.
Delay comparison misses the best immediate option	Construct a small case where waiting loses value despite an informative result; compare full policies from now.	Revise the objective if the isolated test confirms a mismatch. This is not yet a confirmed defect.
Apparent history is insufficient	Fit real preceding-session features and compare with authored, population and demographic priors.	Retain only personalization supported by temporal evaluation.

Each test needs an expected pattern under the competing explanation, not simply an assertion that a change improved the mean. Keep every case, seed, input manifest and selected observation. Analyze tail errors and confidently wrong comparisons alongside overall results. [L03] [L05]

Current delayed-value code motivates a focused research question: using delayed-action losses on both sides can answer a different question from whether waiting beats the best immediate action. Code inspection alone does not establish the size or direction of any resulting error.

05 / REFERENCE REPOSITORIES

Repository comparison: substance and limits

Selected code, model, data, evaluation and license files were inspected; external projects were not executed.

Reference	Verified strength	Limit / decision
digital-patient	Graph-based forecasting and a published research architecture.	Strongest methodological inspiration; simulated cases and example evaluation need adaptation. [R23] [R24]
LifestyleCorp	Broad modular structure and visual ambition.	Explicitly incomplete; inspected simulation service empty. Do not adopt as a working physiological foundation. [R26] [R27]
BioTwin AI	Substantial application and review workflow structure.	Inspected service uses authored numeric formulas; clinical calibration is not established. [R29]
ARYANJUMDE GlucoTwin	Real feature, training and model-comparison pipeline.	Random row evaluation and test-set model selection leave generalization questions. [R30]
Amity GlucoTwin	Training artifacts, transparent limitations and submission materials.	Synthetic random-row metrics do not establish unseen-patient performance. [R33] [R35]
Anurag ICU project	Monitoring UI and model-integration contract.	Prediction page says integration pending; not evidence of an operational CNN-BiLSTM predictor. [R37]
UMCU sasicu	License reviewed only for adoption decision.	Prior permission required, including AI input. Implementation analysis stopped; no adoption planned. [R38]

The two GlucoTwin endpoints differ: >=40 mg/dL rise anywhere within two hours versus >=180 mg/dL exactly two hours later. Different labels, prevalence, synthetic generators and splits prevent direct accuracy ranking. [R31] [R34]

Borrow useful principles and credit sources. A larger feature list, repository name, architecture diagram or high synthetic score is not evidence of better clinical performance.

05 / REFERENCE REPOSITORIES

How digital-patient could improve RENALIS

Use its architecture as a research hypothesis, with clear adaptation and rejection criteria.

The paper combines mathematical physiology with graph forecasting and generative integration and evaluates two simulated cases. Its useful architectural principle is a composable representation of linked processes. It does not supply a validated dialysis model or establish superiority over our current estimator. [R23]

Adopt / investigate	Why	Verification before use
Structured model graph	Expose fluid, refill, cardiac and vascular assumptions and observation links.	Every node/edge maps to actual computation and data lineage.
Modular transition interfaces	Swap physiological components without rewriting the interface.	Shared units, state contracts, conservation and identical starting conditions.
Graph forecasting candidate	Potentially capture temporal relations and provide a surrogate for expensive simulation.	Compare with simple temporal and physiological baselines on identical held-out tasks.
Separate calibration layer	Evaluate honest factual forecast ranges.	Held-out coverage, width and subgroup/shift behavior; no treatment-effect guarantee.

The example repeats simulated trajectories, scales before splitting and shuffles overlapping windows. Those choices are not evidence of unseen-patient generalization. The prediction implementation includes a processing cap; graph-attention dependencies and old interfaces need modernization. Reusing the source would require adaptation and retained Apache notices. [R24] [R25]

The adoption decision should be reversible

Keep a learned graph component only if it improves a predeclared factual objective, calibration or computational efficiency without unacceptable loss of reliability. A transparent computational graph remains useful even if a GNN does not beat a simple baseline. Attention weights alone should not be displayed as proof of physiological causation. [R43]

05 / REFERENCE REPOSITORIES

What to borrow, what to redesign

Adopt inspectable patterns; track source provenance at the level of code and assets.

Reference lesson	RENALIS response
BioTwin: review workflow	Record clinician/researcher notes, original and revised alternatives and rationale. Do not convert formula-derived confidence into a validated probability. [R29]
ARYAN: integrated training pipeline	Provide one reproducible path from permitted data to features, model, evaluation and application artifact. Use separate model-selection and final-test sets. [R30]
ARYAN: what-if feature edits	Label feature sensitivity honestly. Re-scoring edited inputs alone does not validate a treatment effect. [R31]
Amity: artifact completeness	Release architecture, presentation, model details, limitations and demo links together. A listed artifact must also be opened and checked. [R33]
Anurag: explicit integration status	Show disconnected, stale, missing-model and unavailable-output states clearly. Do not let a polished model label imply integration. [R37]
Lifestyle: broad visualization	Use original relevant visuals tied to actual states. Avoid importing empty foundations or unverified asset rights. [R26] [R28]

Reuse and licensing register

digital-patient includes Apache-2.0 terms. The nested ARYAN GlucoTwin and Amity projects contain MIT licenses. Retain the relevant notices for adopted portions and document modifications. LifestyleCorp's inspected LICENSE is empty despite its README's MIT statement. Other missing grants and third-party assets need clarification before copying. UMCU implementation remains excluded under its custom terms. [R25] [R28] [R32] [R36] [R38]

Proposed register fields: source URL, inspected commit, component, purpose, copied versus inspired, license, retained notices, modifications, asset rights and verification date. Public source visibility is separate from permission to reuse every component.

06 / TARGET DESIGN

Decision record: retain or pivot?

The recommendation follows the task and available evidence, not sunk cost.

Alternative	Potential benefit	Main weakness	Decision
Whole-body platform	Broad anatomy and long-term extensibility.	Outside localized challenge scope; the supplied reference is incomplete; far more unobserved physiology.	Reject as the hackathon core.
Glucose-spike twin	Clear wearable stream and established benchmark pattern.	Several close reference projects; would require new domain evidence without a demonstrated advantage.	Retain as an alternative only if dialysis data feasibility fails.
Generic ICU twin	Rich monitoring and many possible endpoints.	Easily becomes a broad dashboard; supplied prediction integration is pending.	Do not pivot on UI breadth alone.
Graph-informed dialysis twin	Fits the condition, current problem and strongest research reference.	Needs validated data semantics, real history learning and policy evidence.	Recommended direction with explicit gates.

The design should become more ambitious within the clinical question: multiple sessions, model uncertainty, resource-aware observation, independent experiments and a path to paired physiology. Keeping one condition does not imply a small technical plan. [R01] [R23]

What evidence could change this decision?

- A clinical collaborator establishes that the proposed decision is not useful or safely testable in its intended workflow.
- Data audits show that the factual endpoint or meaningful historical fusion cannot be supported with permitted inputs.
- An alternative domain offers substantially stronger available data and a demonstrably clearer, distinctive user need.
- Independent evaluation repeatedly shows that the uncertainty workflow is misleading and cannot be corrected or usefully bounded.

Decision owner and review: project lead with clinical/domain review for clinical assumptions. Revisit after the data audit and first historical-personalization evaluation, rather than automatically after adding a new framework.

06 / TARGET DESIGN

Target architecture and evidence boundaries

A modular system in which every displayed claim has a traceable source.

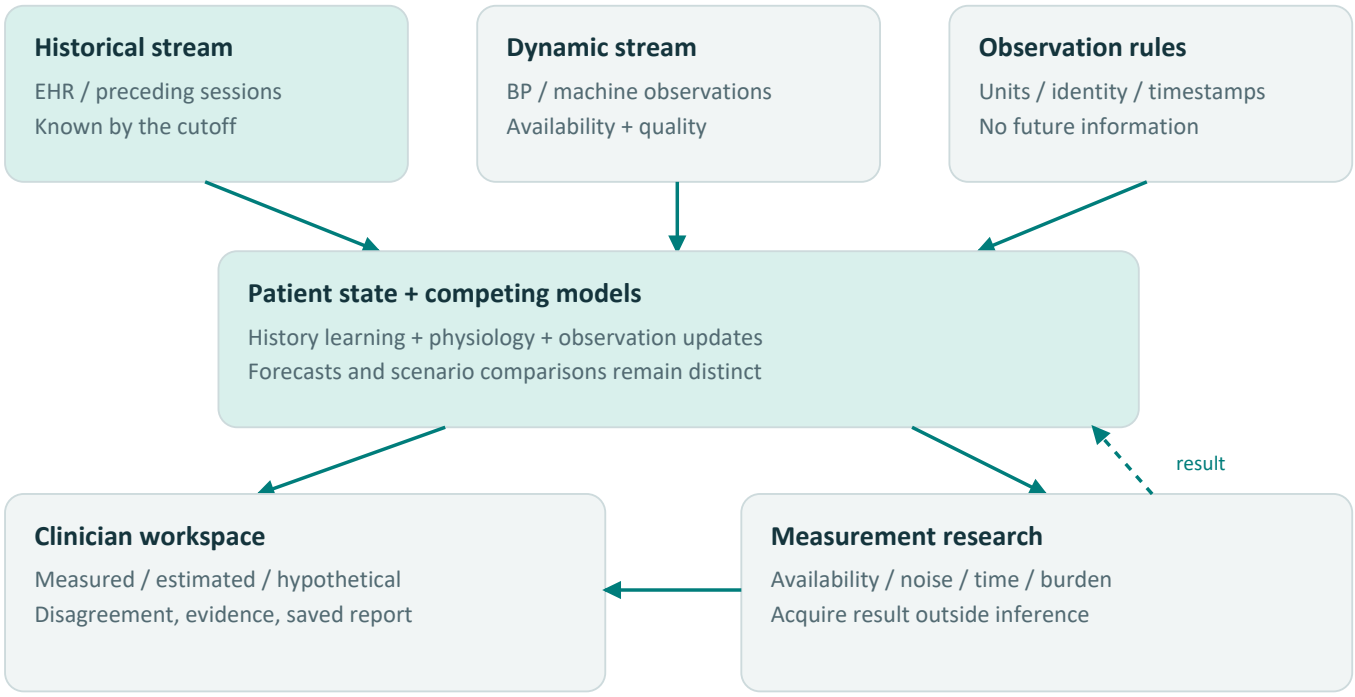


Figure 4. Proposed architecture. The observation/acquisition process must remain isolated from model selection, and retrospective prediction and hypothetical-response validation remain distinct evidence tracks.

Proposed module	Contract and retained evidence
Data adapter + availability ledger	Canonical units, patient/session identity, observed_at and available_at, source, quality, corrections and treatment provenance.
History learner + state estimator	Completed history produces a prior/representation; current prefixes update state without future access. Save the fit window and uncertainty.
Model bank + scenario evaluator	Explicit equations, physiological bounds and observation maps. Alternatives share state and action-time conventions.
Observation policy	Candidate channel/time, predictive distribution, failure/delay assumptions, expected value and reason for abstention.
Evidence API + UI + audit	Stable result schema with observed/estimated/hypothetical status, model revision, assumptions and immutable saved reports.

FastAPI/NumPy and React/TypeScript can remain the core. Add complexity for a demonstrated contract or performance need: model registry, event store, external adapters or an experiment scheduler. No model should silently consume unreleased observations through a shared fixture or UI state. [L05]

06 / TARGET DESIGN

A physiology graph people can inspect

A proposed computational representation, not a clinically identified causal graph.

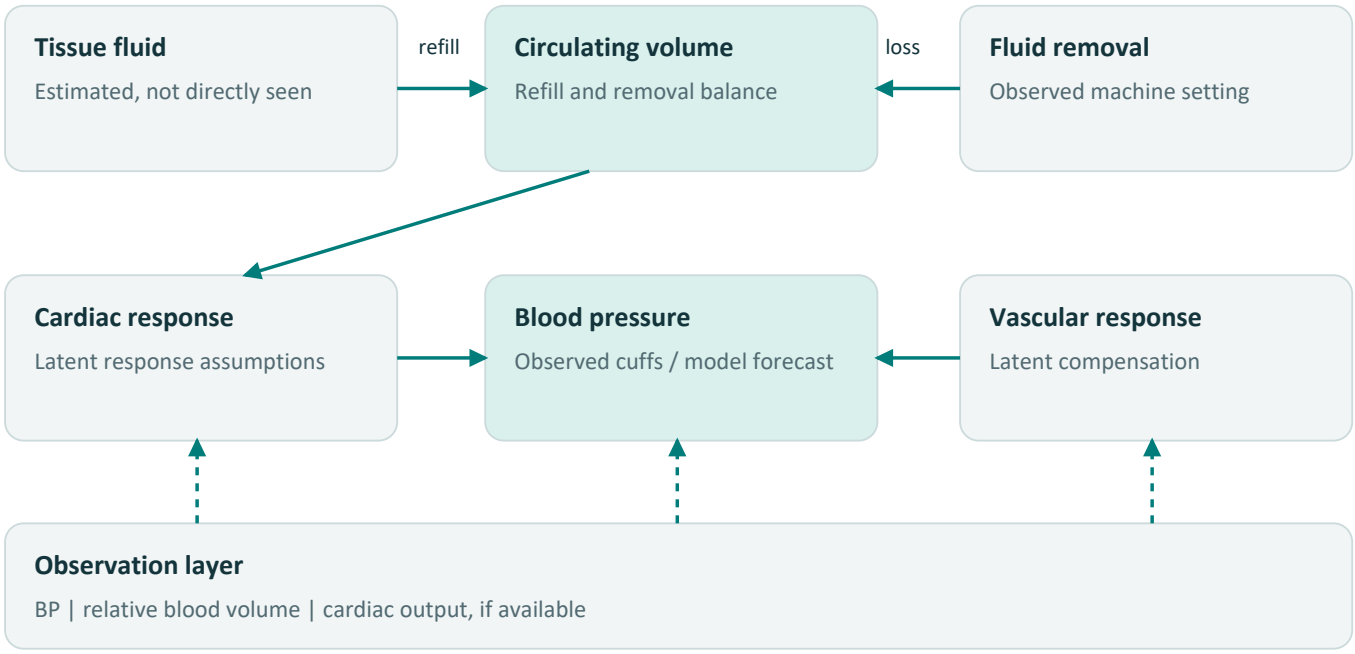


Figure 5. Proposed model organization. Solid links indicate modeled transitions or response dependencies; dashed links indicate observation constraints. Directions and equations must be explicitly specified and validated. This diagram is not a treatment protocol.

Every state node should show a value or distribution, units, timestamp, provenance, physiological bounds and whether it is directly measured or inferred. Every edge should reveal its equation or response assumption, parameter uncertainty and supporting rationale. Selecting an explanation highlights the assumptions that differ. [R03] [R23]

Three layers that should remain distinct

- Mechanistic state layer: mass-balanced fluid transitions and explicitly authored response relations.
- Observation layer: channel-specific mappings, quality, bias and uncertainty; missing measurements remain missing.
- Optional learned layer: a temporal residual or graph surrogate evaluated separately for factual forecasting or speed.

Do not estimate every latent physiological quantity from BP alone. When multiple latent states explain the available data, preserve a broad or multi-modal estimate and expose its consequences. A faithful graph can clarify assumptions even when the underlying physiology is unidentifiable.

06 / TARGET DESIGN

Make historical personalization real

Replace decorative patient history with temporally valid algorithmic influence.

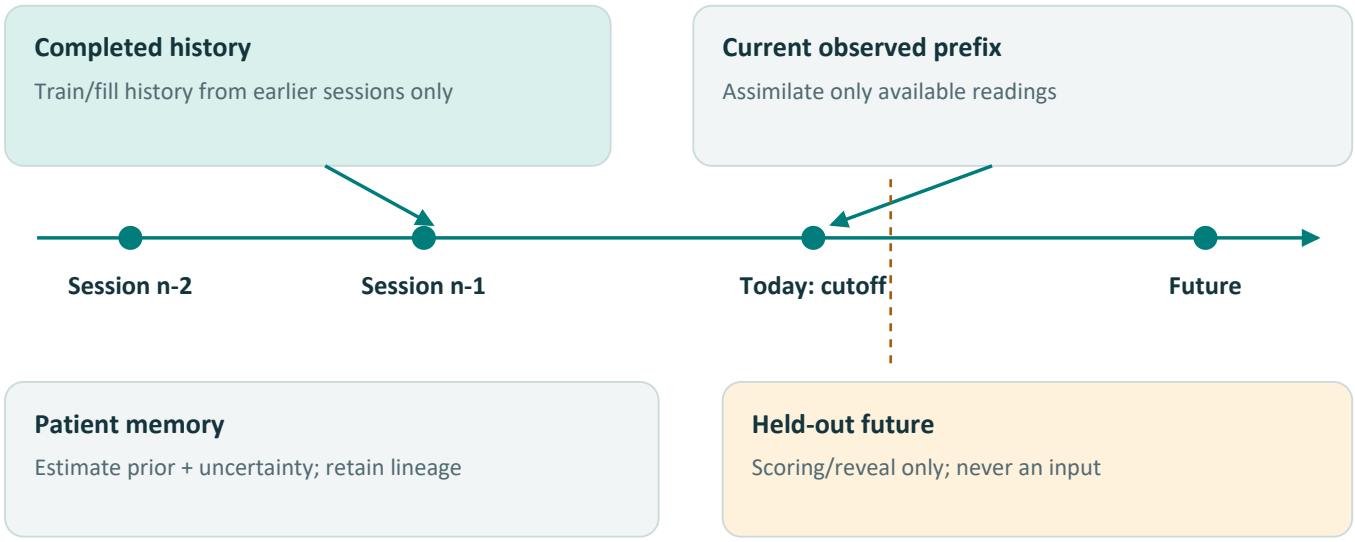


Figure 6. Temporal contract for personalization. Preceding completed sessions can inform today's prior; current observations update it. Future current-session outcomes are reserved for scoring and reveal.

Candidate history	Eligibility rule	Test of value
Prior nadirs and event history	Only completed sessions known before today's cutoff; distinguish first/recurrent episodes.	Does this beat demographic and prior-occurrence baselines?
Observed prior response to settings	Keep action timestamps, confounding context and missingness; do not infer causal benefit from association.	Does history improve factual forecasts on later sessions?
Weight / prescribed target / interval	Audit event time and record availability; current session-end weight cannot leak backward.	Are errors reduced on identical windows with uncertainty?
Learned prior or patient representation	Estimate only supportable dimensions; retain uncertainty with few or poor-quality sessions.	Cold-start, short-history and longer-history comparisons.

Use both unseen-patient evaluation and forward evaluation for known patients with preceding history; these answer different questions. Group calibration and final test sets separately. Audit HEMOBP source-unit conversion, session identity and informative sampling before fitting richer history. [R04] [L02]

A patient-specific predictive representation can be useful without claiming that every latent parameter has been physiologically identified. Report incremental value and its uncertainty; do not rename a demographic feature block as learned longitudinal physiology.

06 / TARGET DESIGN

Competing explanations, shared evidence

Close factual fit can coexist with disagreement about a hypothetical response.

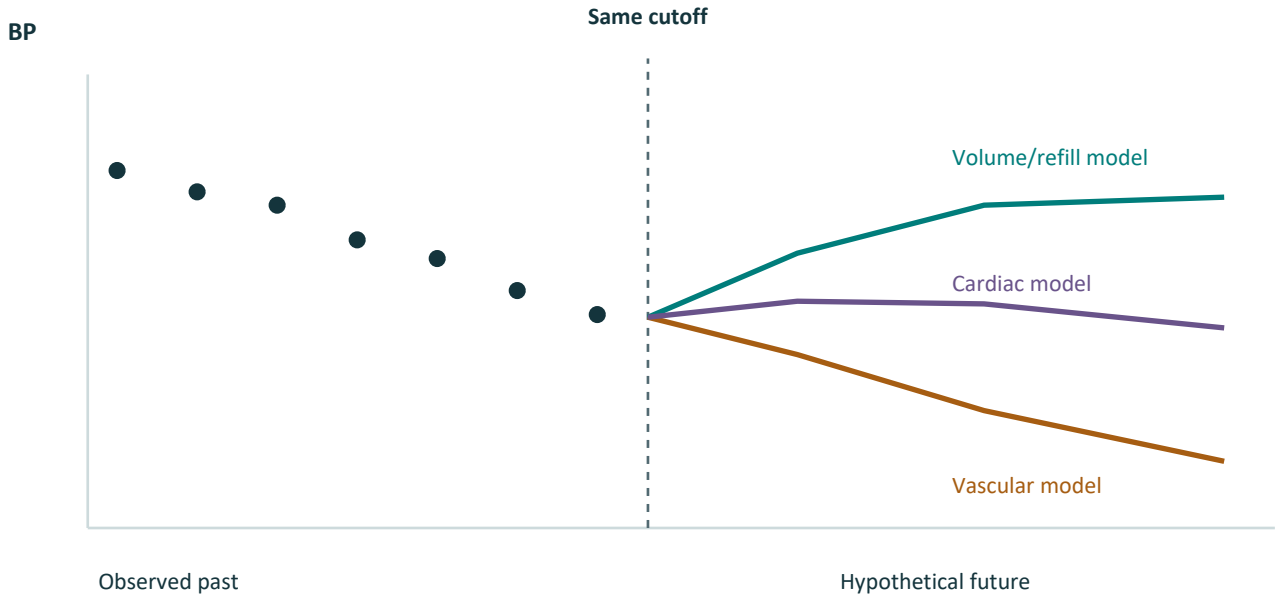


Figure 7. Schematic illustration, not measured RENALIS trajectories or clinical effect estimates. Models share the observed prefix and starting cutoff, then differ under the same specified hypothetical profile.

The interface should first show that models explain the same available past. It should then identify which assumptions produce different response estimates. Relative compatibility weights are conditional on this model set; a clinically missing mechanism can make every family misleading. [R03] [R13] [L04]

Proposed capability	Acceptance condition
Mixed mechanisms	Synthetic mixtures do not force a pure-family winner; report overlapping support and scenario consequences.
Unknown mechanism state	Out-of-family observations can trigger reduced authority; quantify false downgrades and detection delay.
Alternative comparisons	All alternatives share the same posterior starting state, horizon and correct action start time.
Support view	Keep factual fit, input-range support, model dependence and causal identification as separate dimensions.
Useful unresolved state	Explain what remains uncertain and which available evidence, if any, could help.

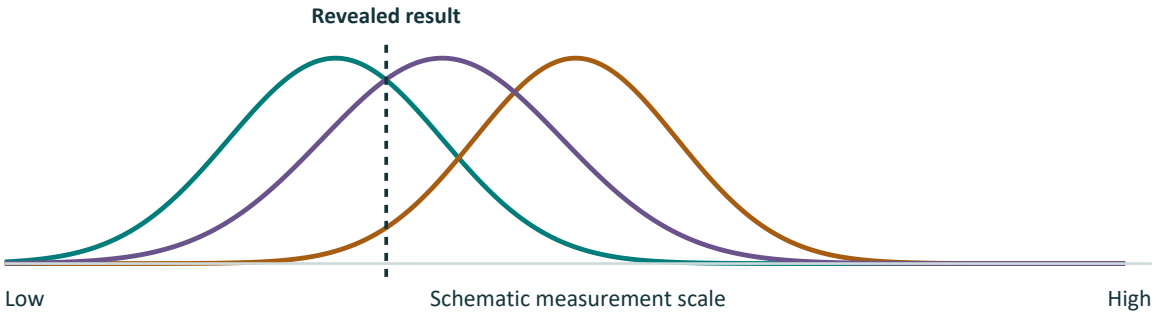
A model may forecast BP adequately but rank hypothetical profiles poorly. Evaluate those tasks separately. Prevent attractive anatomical animation from turning an uncertain model interpretation into an apparent diagnosis.

06 / TARGET DESIGN

What would change our mind?

Make each observation experiment falsifiable before revealing its result.

BEFORE: predicted observation distributions



Volume/refill | Vascular | Cardiac

Fits several models
Retain uncertainty
Do not force a winner

Fits one better
Update compatibility
Recheck scenario ranking

Fits none well
Flag model inadequacy
Withhold comparison

Figure 8. Illustrative normalized distributions only; no device units or patient risk are implied. Show competing predictions before acquisition, then overlay the available result and retain all relevant outcomes.

Question on the measurement card	Required explanation
Why this channel?	Which models predict different readings and why that difference could affect the comparison.
What might it show?	Overlapping predictive ranges with units and explicit model/noise assumptions.
What would change?	Possible changes in compatibility and scenario ranking; distinction is not guaranteed.
What if it conflicts?	Check quality and record corrections; do not force a winner when every model fits poorly.
What remains unresolved?	Residual uncertainty, unavailable channels, ambiguity and limits of hypothetical responses.

The Meera minute-90 example already illustrates a useful non-resolution: the synthetic 3.74 L/min result weakens one family but leaves volume/cardiac ambiguity. Retain that honest outcome and add examples of no value, conflict and conditional agreement. The channel model and value are authored research elements, not clinical device validation. [L04]

Verification: the selector cannot access unreleased outcomes; distributions are computed from the current model state; acquisitions become available at their stated time; duplicates are idempotent; result updates and saved reports retain complete lineage.

06 / TARGET DESIGN

Value of information includes time

Optimize a declared research objective over complete policies from the present.



A useful result can still arrive too late. Acquisition is an evaluated choice.

Figure 9. Proposed policy comparison. Score the best immediate no-measurement option and the complete acquisition path over the same time horizon, with consistent loss accounting.

The research definition should be written before implementation: net observation value is loss under the best immediate option minus expected loss under an observation-and-response policy, minus acquisition burden. Waiting consequences belong in the trajectory; avoid counting the same waiting loss twice. A failed or late result still incurs its appropriate burden.

Quantities that need independent definitions

- Information gain: reduction in model-family uncertainty. It can be high even when the scenario choice does not change.
- Decision value: expected improvement under a declared loss, conditional on model and observation assumptions.
- Clinical utility: a clinically meaningful benefit/cost tradeoff that needs domain agreement and appropriate empirical evaluation.

The existing research loss combines threshold crossing, below-threshold severity and unfinished fluid target, with authored burden/delay terms. These terms are inspectable, not validated utilities. Start with sensitivity analysis, compare quadrature to a numerical reference and test action timing independently. Preserve no-acquisition as a valid outcome. [L04] [L05]

A useful falsification case: an accurate result arrives only after the relevant opportunity has passed. The method should not prefer it merely because it makes model identity clearer.

06 / TARGET DESIGN

A practical observation workflow

Availability and acquisition quality are part of the model, not just disabled buttons.

Authored test configuration	Inputs / capabilities	Required behavior
Basic unit	Intermittent cuff BP, machine settings and routine/manual observations.	Show what can be inferred, when another cuff helps and what cannot be distinguished.
Equipped unit	Basic configuration plus an available RBV channel.	Use quality, delay and failure assumptions; do not automatically prefer the extra sensor.
Research setting	Additional appropriate hemodynamic measurements with operator/time constraints.	Represent method-specific uncertainty and feasibility; keep experiments explicitly synthetic or appropriately governed.

These are proposed scenarios, not estimates of Indian equipment prevalence. Clinical collaborators should specify real acquisition processes, staff roles, delays and failure conditions. Published practice variation motivates that study, while device-specific validation prevents treating all output or volume channels as equally reliable. [R41] [R42]

Data contract for an observation request

Record channel, units, requested_at, observed_at, available_at, source device or manual operator, quality, status, failure reason, correction history and acquisition burden. Predictions use availability time; clinical displays can also show observation time. A late correction must not rewrite what the algorithm knew in an earlier saved report.

Workflow completion

A nurse/researcher can record acquisition and quality; the clinician/reviewer can inspect what changed, document a judgment and produce a concise handover. Any experimental comparison remains distinct from a treatment command. Missing hardware should yield an understandable unresolved state rather than a fabricated substitute measurement.

06 / TARGET DESIGN

One screen, one patient, one question

Reorganize the interface around comprehension and traceability.

Patient + one time cutoff

Session | data freshness | provenance | review status

What is known / what may happen

Observed BP + machine context
Recorded symptoms and data quality
Exact endpoint + conditional forecast

Why is this uncertain?

Shared history, competing explanations
Different hypothetical responses
Assumptions and unavailable evidence

What observation could help?

Expected results, availability and delay
Acquisition cost, failure and no-value state

What changed / can I audit it?

Before/after comparison and uncertainty
Saved evidence, model revision and rationale

Figure 10. Proposed information architecture, not a screenshot or an implemented redesign. It preserves a single patient/time context and separates knowledge, forecasts, uncertainty, acquisition and review.

Use consistent visual encodings for observed, inferred and hypothetical quantities across the landing page, workspace, graph and saved report. Every chart needs units, cutoff, horizon and data freshness. The observed benchmark should remain a separate evidence view with its own task definition. [L04]

Interaction	Purpose / acceptance
Click a model node	See inputs, equation, uncertainty, evidence and what is not measured.
Switch explanations	Keep the same past and cutoff; highlight changed assumptions and scenario effects.
Inspect a measurement	Read predicted ranges and feasibility before seeing the result.
Inspect the update	See before/after compatibility and scenario comparison, including remaining ambiguity.
Save and revisit	Reopen the original evidence and revision; recomputation is explicitly a new result.

Use progressive disclosure for numerical detail. Primary copy should explain the clinician's question in connected prose. Keyboard navigation, labels, contrast and responsive charts belong in acceptance checks; a visually polished screen is not yet evidence of clinical usability.

07 / VALIDATION SCENARIOS

Realistic scenarios: personalization and doubt

Each scenario needs inputs, expected behavior, a failure condition and retained evidence.

Scenario	What the reviewer sees	What must be tested
Same current BP, different histories	Two patients share a current reading but have different genuine preceding-session records. The UI identifies which historical inputs changed the prior or prediction.	No future access; same scoring task; quantify incremental value and uncertainty. Changing a decorative demographic should not falsely imply learned physiology.
Ambiguous history, diverging futures	Several models fit the past comparably but disagree under the same removal profile. The shared prefix remains visible.	Model fit, parameter bounds, scenario ranking and false certainty. A well-fitting family is not automatically the true mechanism.
Useful observation, incomplete resolution	Predicted channel ranges are shown before acquisition. The available result weakens one explanation but leaves two plausible.	All updates use declared likelihoods and timestamps. The interface must keep the unresolved comparison rather than celebrate forced certainty.
Informative but decision-irrelevant	A measurement sharply separates families, yet all supported families prefer the same comparison.	Information gain and decision value differ. The policy should not claim decision benefit merely from a lower entropy.

Use deterministic synthetic cases for demonstration and independent cases for aggregate testing. A memorable patient story is a communication asset, not a sample-size argument. Present the observation as synthetic where appropriate and keep unchosen treatment outcomes hypothetical.

Acceptance is not always a resolved answer. For some cases the intended successful behavior is a clear explanation that the available evidence remains insufficient.

07 / VALIDATION SCENARIOS

Realistic scenarios: operational constraints

The workflow should remain understandable when conditions are inconvenient.

Scenario	Expected behavior	Failure to detect
No auxiliary hardware	List only feasible channels, explain what remains unknowable and allow no acquisition.	Invented device data or an impossible measurement recommendation.
Late or failed acquisition	Retain request and burden; advance time correctly; update only after availability.	Retroactive action change, erased failed requests or missing waiting consequences.
Current low BP / symptoms	Label an existing threshold event separately from advance warning; recorded symptoms remain visible above the research threshold.	Displaying a future-warning percentage for an already present event or dismissing recorded concern.
Artifact / stale / correction	Show quality and freshness, withhold unsupported comparison, preserve correction lineage.	A stale normal reading implies stability or a correction rewrites an earlier saved result.
Recovery and recurrence	Mark prior events and recovery; score first-event and recurrent tasks separately.	A recovered current BP is treated as proof of no earlier event.
Stable or completed session	Avoid unnecessary acquisition; session completion ends treatment forecast and keeps the report.	Persistent invented warnings or projections beyond a completed treatment horizon.

The current project already contains several safeguards. The proposed work is to unify their meaning across workspaces, test more realistic acquisition paths and measure user understanding. Clinical response protocols and thresholds require domain review; this report does not prescribe bedside action. [L04]

Add explicit scenarios for clinician action, interrupted sessions, missing follow-up and an observation contradicting every family. Each should have a machine-readable fixture plus a plain-language explanation of the expected result.

08 / EVALUATION PROTOCOL

Evaluation I: factual forecasting

Specify the estimand and split before fitting, then compare identical eligible examples.

Define the task and observation process

Predeclare population, cutoff, threshold inequality, horizon, permissible inputs, missing-follow-up handling and event episodes. Distinguish a future measured low-BP event from unobserved continuous hypotension and symptomatic IDH. Use separate first-event and recurrent-event analyses. Define treatment changes and observation schedules as part of the data-generating process. [R11] [R12]

Design element	Proposed protocol / artifact
Data audit	Source checksums, inclusion flow, measurement-ID rules, session audit, unit conversions, time/availability dictionary and future-input exclusions.
Generalization	Patient-disjoint training/calibration/test sets; additionally forward-session testing for personalized known-patient use. Retain split manifests.
Comparators	Last value, trend, prior-occurrence, dynamic-only, demographics, preceding history, physiological estimator and justified learned candidates.
Model selection	Tune on development/validation only; lock final threshold and model before testing. A final test is not a model-selection leaderboard.
Forecast metrics	MAE and bias; range coverage and width; discrimination; calibration/Brier or log loss where applicable; event sensitivity, PPV, lead time and alerts per session/hour.
Uncertainty	Paired patient/session-block intervals and subgroup counts. Correlated windows are not independent people.

Do not require an arbitrary improvement target invented for a pitch. Predeclare the intended decision context and a meaningful effect or precision target with collaborators. Publish negative or small increments, missingness, exclusions and limitations. TRIPOD+AI can structure reporting. [R17]

Prediction calibration is a separate evaluation from error. A conformal method may be investigated with explicit assumptions and shift tests; it does not automatically provide per-patient guarantees or validate hypothetical treatment effects. [R43]

08 / EVALUATION PROTOCOL

Evaluation II: mechanisms and scenarios

Separate observational fit, parameter plausibility and hypothetical-response performance.

Research question	Experiment	Required reporting
Can the states be identified?	Use known synthetic states and observation schedules; vary parameter degeneracy and channel availability.	Identifiable regions, broad/multi-modal estimates and failure cases.
Does model diversity help?	Compare single-family, averaged and explicit-disagreement versions under the same histories.	Factual error, scenario ranking, false certainty and abstention.
Does it handle missing physiology?	Independent mixtures, nonlinear response, drift and omitted mechanisms.	Confident errors, contradiction delay, false downgrades and unresolved duration.
Are parameters plausible?	Bounds, conservation checks, residual analysis and sensitivity to priors.	Fits obtained only through implausible states are surfaced or rejected.
Are alternate effects identified?	Separate in-silico scoring from any later causal or experimental response study.	A factual future reveals one path, not the unchosen alternative.

Create an independent simulator specification and review its assumptions before generating a holdout. Independence is not just another random seed in the same pressure equation. Include outcome dynamics and sensor relations absent from the estimator, while keeping problems physically interpretable. Publish the common loss and its limitations.

For future observed treatment-response analysis, specify intervention versions, confounders, positivity and the longitudinal causal estimand. Clinician actions can depend on earlier observations; naive before/after comparisons can be confounded. If the assumptions are unsupported, keep alternate-profile effects experimental. [R15]

A fitted explanation is a model interpretation. A correct factual forecast is a prediction result. A validated intervention response is a different claim requiring different evidence.

08 / EVALUATION PROTOCOL

Evaluation III: observation policies

Use equal resource rules and score complete policies, including choosing not to measure.

Protocol component	Proposed specification
Baselines	No additional observation; fixed cuff; fixed auxiliary channel where available; fixed schedule; information gain; decision value; reference/oracle bounds.
Fairness	Same histories, observation draws, channel availability, maximum budget, timing and failure rules. Charge requested failed acquisitions consistently.
Outcomes	Independent decision regret; worst/tail regret; cases worse than baseline; acquisition burden; failures; delay; unresolved comparisons; false certainty.
Numerics	Compare low-order quadrature with high-order or Monte Carlo references; retain value error and candidate-rank stability.
Robustness	Noise/bias, channel correlation, unknown mechanisms, priors, support cutoffs, loss weights, missingness and availability shifts.
Evaluation integrity	Current 108 cases remain diagnostic; separate development and sealed holdout families/seeds; report every scored case.

Compare the entire cost-performance frontier. A policy that acquires less may have useful burden tradeoffs even if mean regret is not lower; that tradeoff must be shown explicitly rather than called unconditional superiority. Separate average results by mechanism class, equipment configuration and waiting time.

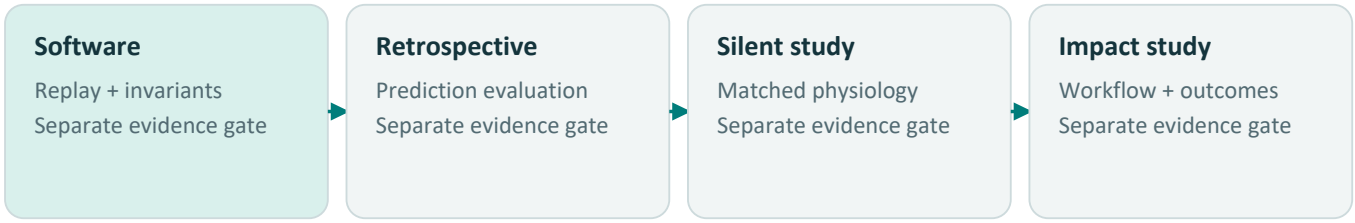
Use shared-history blocks for paired uncertainty in synthetic evaluation and patient/session clusters for observed work. Include a no-information case, an always-unresolved case and a case where the acquired value makes a particular decision worse. Expected value is not a promise about every realized result. [L03]

Clinical acceptability and utility weights require clinical agreement and appropriate evidence. Authored research units are useful for testing the algorithm, but should not be relabeled as measured staff costs or patient benefit.

08 / EVALUATION PROTOCOL

A credible path to clinical evidence

Ambitious translation through distinct studies and explicit claim boundaries.



Claims expand only when the corresponding evidence exists

Synthetic behavior does not establish patient benefit.
Observed prediction does not identify an unobserved treatment effect.
Clinical experts and institutional governance define later studies.

Figure 11. Proposed evidence progression. These are project gates, not regulatory classifications or guarantees of approval.

Stage	Data / process	What it can establish
Expert workflow study	Scenario walkthroughs with clinicians, nurses and intended reviewers; document misunderstandings.	Usability, relevance and realistic acquisition assumptions; not clinical efficacy.
Silent paired physiology	Appropriately approved observational collection of synchronized BP, machine settings, symptoms and available volume/hemodynamic channels.	Observation models, feasibility, missingness, failure and silent predictive performance.
External validation	Multiple sites, devices and patient groups; predeclared transportability analysis.	How well predictions and uncertainty transfer to the tested settings.
Clinical impact study	Clinician-led protocol, governance, endpoint selection and sample-size design.	Effects of the whole workflow on predeclared outcomes, subject to study design.

Choose sample size from precision, event rates, clustering and the intended comparison rather than selecting a convenient window count. Later clinical interaction needs ethics/institutional review and the right domain partners. DECIDE-AI is a useful reporting/design checkpoint for early clinical evaluation. [R18]

No clinical data collection, patient intervention or external outreach is performed by this report. Scientific stages remain in the plan; authorization to create a PDF is not authorization to conduct a clinical study.

09 / MASTER PLAN

The full roadmap is dependency-driven

A substantial program of work, with a complete release at each evidence stage.

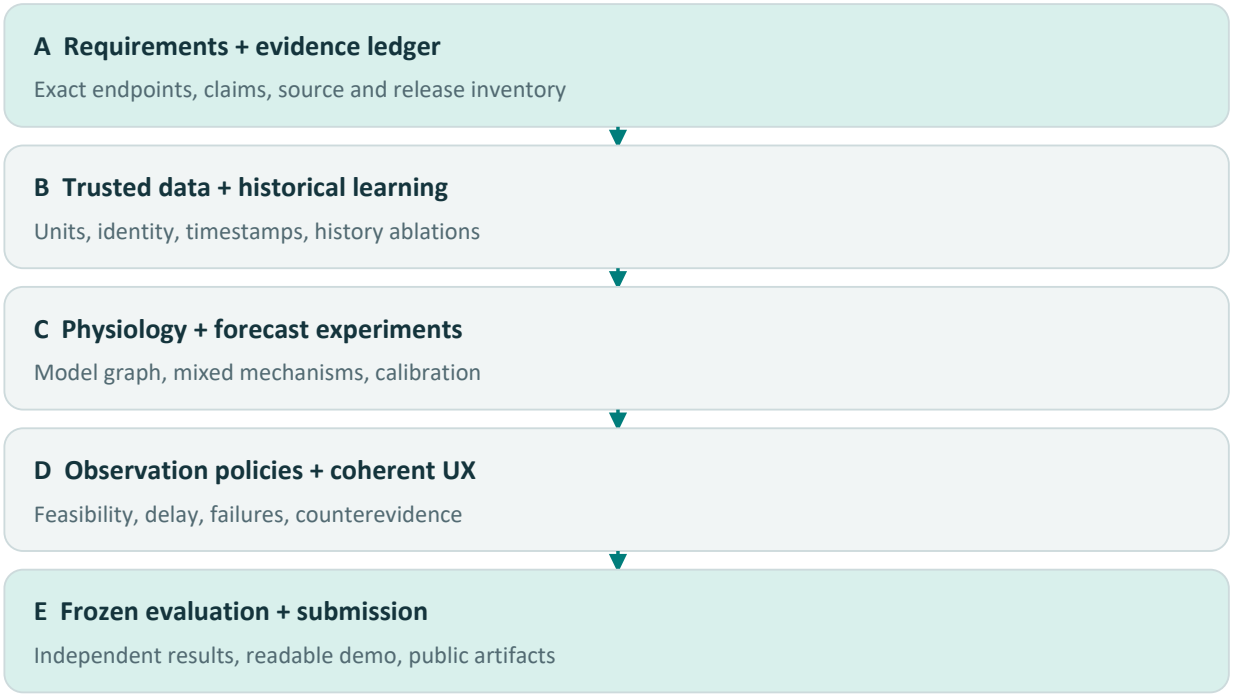


Figure 12. Core dependency path. Clinical discovery, communication, usability and publication preparation can inform each stage; data integrity and evidence gates cannot be bypassed.

Release / milestone	Concrete completion condition
R1: trusted research foundation	Audited task/data definitions, source/claim ledger, reproducible existing results and explicit known failures.
R2: personalized explanatory twin	History learning, faithful physiological graph, mixed/unknown model behavior and identical-cutoff comparisons.
R3: feasible evidence acquisition	Time/failure/resource-aware observation policies with independent tests and understandable updates.
R4: externally reviewable package	Frozen results, coherent clinician walkthrough, public-ready repository, architecture/deck and full demo.
R5+: translation program	Matched physiology, multi-centre transportability, silent studies and appropriately governed impact evaluation.

This roadmap does not assume a small number of builders and is not an excuse to omit ambition. It distinguishes the dependency order from external competition deadlines. If a submission is due before later studies are possible, the release should accurately present completed evidence and the larger plan.

09 / MASTER PLAN

Work packages: data and patient memory

Each work package has an artifact, a dependency and an observable exit condition.

Package	Deliverable and dependency	Exit condition
WP01: requirement / claim ledger	One authoritative mapping from brief/form to artifact and model claim. Depends on supplied requirements and final project scope.	Every mandatory item and public claim has a source, owner/status and verification step.
WP02: cohort and units audit	Data dictionary, sample audit, measurement identity, session reconciliation, canonical conversions and exclusions.	Counts reconcile; future fields are unavailable to earlier forecasts; unresolved unit/session issues are reported.
WP03: event semantics	First/recurrent event episodes, recovery, measured horizon and missing-follow-up rules. Depends on WP02.	Independent hand-checked examples match the labeler; prediction and current-event states are distinct.
WP04: history features	Preceding-session features and availability ledger. Depends on WP02-03.	Each feature has source/time lineage and no current-session outcome leakage.
WP05: history learner	Population, demographic and history-informed priors/representations, with cold-start handling.	Temporal ablation on identical examples, paired uncertainty and explicit unsupported latent dimensions.
WP06: factual model benchmark	Simple baselines and justified learned candidates with calibration and final-test separation.	A reproducible report includes performance, workload, calibration, subgroup counts and all exclusions.

Possible advanced candidates include hierarchical patient effects, state-space learners, temporal neural models and a graph residual/surrogate. Model choice follows the task and held-out result. Maintain a simple baseline throughout; architectural complexity is not itself an exit condition.

Proposed artifacts: endpoint-specification.md, data-dictionary, cohort-flow, split-manifest, availability audit, history ablation report, calibration report and model cards. These names are plans, not newly created deliverables.

09 / MASTER PLAN

Work packages: physiology and acquisition

Build the explanatory and observation layers around shared contracts and falsifiable assumptions.

Package	Deliverable and dependency	Exit condition
WP07: state/graph contracts	Explicit state, transition, observation, units and parameter interfaces. Uses WP02 and current equations.	The explanatory graph matches computation; conservation, limits and deterministic replay pass.
WP08: model diversity	Mixed mechanisms, omitted-model cases and transparent contradiction handling. Uses WP07.	False certainty, abstention and detection behavior are reported on independent cases.
WP09: acquisition ledger	Requested/observed/available times, quality, failure, correction and burden. Uses WP02 and equipment scenarios.	Delayed, duplicate, failed and corrected observations preserve immutable evidence and time integrity.
WP10: policy objective	Whole-policy comparison from now, same horizon, declared costs and no-acquisition option. Uses WP07-09.	Hand-checkable examples confirm waiting/failure accounting and comparisons with best immediate action.
WP11: numerical policy engine	Predictive distributions, value integration and accuracy reference. Uses WP10.	Integration error and candidate ordering are checked against a numerical reference.
WP12: independent challenge	Frozen diagnostic, development and new holdout suites; equal observation rules.	Complete paired results show mean/tail regret, adverse cases, acquisition tradeoffs and uncertainty.

Potential research extensions include multi-step acquisition, a limited measurement budget across a session and robust objectives under model uncertainty. Introduce them after the one-step objective is correct; evaluate the additional benefit separately. Active information gathering should not imply permission to perturb a patient or machine.

Proposed artifacts: model/observation specifications, physiological bounds, independent simulator specification, numerical verification report, acquisition failure fixtures, policy comparison report and counterexample library.

09 / MASTER PLAN

Work packages: experience and engineering

A rich product experience with clear scientific boundaries and reproducible execution.

Package	Deliverable and dependency	Exit condition
WP13: unified workspace	One patient/cutoff, readable current state, forecast, explanations and evidence acquisition. Uses stable WP07-11 APIs.	Core scenarios can be completed without losing patient/time context or mixing observed and hypothetical results.
WP14: graph and counterevidence views	Clickable computational graph, predicted measurement ranges and before/after update.	A reviewer can trace a model claim to its inputs and explain what would weaken it.
WP15: audit / handover	Immutable snapshot, corrections, review notes and concise export. Uses WP09.	Original saved reports remain reproducible after resets, new observations and model upgrades.
WP16: operational resilience	Deterministic replay, input validation, recovery, offline demonstration and clear disconnected states.	Failure/retry paths retain provenance; app remains inspectable under tested network and data interruptions.
WP17: usability / accessibility	Structured tasks, readable charts, keyboard flow, contrast and responsive layouts.	Measured task outcomes and misunderstandings drive revisions; all core states are covered.
WP18: experiment/release system	Dependency locks, data/model hashes, versioned evaluations and source/license register.	A clean environment reproduces the allowed synthetic demo and specified reports.

Additional engineering can include event-backed ingestion, a model registry and experiment tracking when they solve a real reproducibility or concurrency problem. Keep the local synthetic experience runnable without paid services. Restrict public artifacts to permitted anonymized/synthetic material and inspect for credentials before release.

Focused regression checks should protect time isolation, event semantics, immutable evidence, acquisition and meaningful numerical behavior. UI verification should test readable outcomes and critical navigation rather than duplicating implementation details.

09 / MASTER PLAN

Work packages: a larger research program

The plan can grow far beyond a submission while keeping the core clinical question intact.

Package / extension	Purpose and evidence gate
WP19: clinical discovery	Document intended decisions, care roles, resource constraints and useful evidence. Structured expert review; no unapproved outreach is performed here.
WP20: matched sensor study	Estimate observation maps, delay, failures, operator time and quality from appropriately governed paired recordings. Sample-size/precision plan before collection.
WP21: multi-session state	Model uncertainty across sessions, interval intake/output and residual function where recorded. Validate transport and conservation; accounting alone is insufficient.
WP22: robust / multi-step sensing	Choose sequences of observations under a session budget and model uncertainty. Compare against fixed and one-step policies on new independent tasks.
WP23: multi-centre transport	Evaluate site/device shifts, subgroup behavior and local recalibration with separate external data. Report failures as well as pooled means.
WP24: silent and impact studies	Progress from silent evaluation to a clinician-led workflow study with ethical/institutional governance and predeclared outcomes.
WP25: interoperable record exchange	Synthetic ABDM FHIR example, validation results and explicit integration limits. Live integration is a separately scoped project. [R19]
WP26: release / submission / dissemination	A coherent public-ready evidence bundle, technical report, concise presentation and complete demonstration. External publication requires authorization.

Long-term research can investigate hierarchical model populations, adaptive recalibration, observation scheduling under shared equipment and multi-session decision support. Each introduces new data and confounding issues. Keep extensions modular and evaluate their incremental contribution rather than turning them into a broader unvalidated diagnosis platform.

Ambition expands the research program. It does not erase the distinction between a plausible plan, a working algorithm, an independent result and demonstrated benefit.

10 / EXPLANATION AND SUBMISSION

The full judged demonstration

Use connected narration, real interaction and graphics that explain the computation.

Time	Story beat	Visible proof
00:00-02:00	Patient problem and localized endpoint	Who uses RENALIS, what is observed and what future event is being predicted.
02:00-05:00	Data fusion and personalization	Preceding history changes a specific state/forecast; show cutoff and unavailable future.
05:00-09:00	A complete primary workflow	Actual patient navigation, measurements, forecast and hypothetical profile comparison.
09:00-12:00	Competing explanations	Same observed history, differing response assumptions and unresolved comparison.
12:00-14:00	The measurement question	Predicted distributions, availability/delay, synthetic acquisition and honest update.
14:00-16:00	Failure and practicality	Unavailable device, contradiction or current-event state; saved handover/evidence.
16:00-18:00	Results and reproducibility	Observed benchmark, negative policy result, frozen evaluation and exact release.
18:00-20:00	Contribution and larger plan	Bounded claim, evidence gaps, clinical-validation path and accessible artifacts.

The existing 2:46 v2 film is a useful introduction. It uses free Kokoro Heart narration, motion graphics and framed product panels. A full submission demo must explain the algorithm, evidence and limitations in sufficient detail, while preserving natural complete sentences and readable resting points. [L06]

Acceptance of the final recording

Inspect every important scene and interaction: narration matches the screen; key numbers match the captured release; full headings/results are visible; captions are synchronized; speech is intelligible; chapter jumps and download/playback work. Confirm actual duration and signed-out unlisted-video access. Do not assume a player link establishes compliance. [L01]

10 / EXPLANATION AND SUBMISSION

Explain the reasoning in sentences

A pitch should connect observations, uncertainty, action and consequence.

The recent video requests asked for a more human voice, more meaningful motion, readable website interactions, more explanatory on-screen material and complete spoken sentences. Preserve that direction. Short phrases work as labels; narration should develop the reasoning and connect each scene to the next. [L06]

Spoken explanation	What must be visible
These readings fit several explanations.	Shared observed prefix and comparable factual fits, with clear labels.
The explanations predict different responses to this profile.	Same hypothetical action and cutoff; divergent model forecasts.
A measurement could help if the models expect different results.	Predicted channel distributions, overlap, timing and feasibility.
The result weakens one explanation but leaves uncertainty.	Revealed value, before/after compatibility and still-unresolved comparison.
Our current policy has not beaten cuff-only mean regret.	Full retained policy table, limitations and the next independent test.

Motion should carry information

Animate an arriving observation, a growing uncertainty range, a highlighted dependency or a model losing compatibility. Hold the screen still while text or a result needs reading. Use actual clicks and entered inputs with visible outcomes. Avoid background scroll that crops a heading while the narration explains it.

Reusable production brief

Inspect the actual release; write a connected story in complete sentences; demonstrate a realistic workflow; give every meaningful spoken idea a corresponding visible explanation; tie animation to real computation; maintain readable safe margins; synchronize to the actual audio; export captions and editable sources; and inspect the final encoded video. A changed voice may require retiming every scene.

This report does not regenerate the video. The communication plan applies the last two relevant requests to the larger judged demonstration rather than replacing the research evidence with marketing.

11 / DECISIONS AND NEXT ACTIONS

Decision ledger: why these recommendations

Recommendations are conditional and revisable; record the evidence that would change them.

Decision	Why it follows	Revisit when
D01 Retain dialysis	Fits localized requirement and coherent physiology/evidence question. [R01]	Clinical usefulness or permitted-data feasibility fails.
D02 Use a model graph	Makes dependencies and uncertainty inspectable; supported by research architecture. [R23]	It does not match computation or users cannot interpret it.
D03 GNN remains a candidate	No verified dialysis generalization advantage is supplied by the reference. [R24]	A fair factual benchmark demonstrates benefit or clear speed value.
D04 Learn real history	Current prior is authored and observed demographic increment is tiny. [L02] [L04]	History has no incremental value or is unavailable; narrow the claim.
D05 Show counterevidence	Uncertainty is meaningful when possible observations can weaken an explanation.	Users overinterpret model ranges as diagnosis or the ranges are uncalibrated without clear labels.
D06 Score whole policies	Delay/failure can undo the value of an informative result.	Isolated tests establish an alternative mathematically coherent objective.
D07 Preserve negative results	Current mean decision regret does not beat cuff. [L03]	A fresh locked evaluation supports a stronger bounded claim.
D08 Mixed/unknown states	Mechanisms can coexist and the model set can be wrong. [R03]	Different model families demonstrate better calibrated and understandable behavior.

A decision record should include the owner, date, evidence, alternatives, consequence, acceptance test and review trigger. A technically impressive replacement is justified by evidence or a clarified product need, not by novelty of its name.

11 / DECISIONS AND NEXT ACTIONS

Risks, open questions and immediate sequence

Resolve dependencies first while keeping the larger plan intact.

Open issue / risk	Next action and completion evidence
Official rubric and winner history unavailable	Retain the inferred review matrix; seek official clarification when possible without presenting guesses as rules.
Video and shortlisting inconsistencies	Use the form/brief ledger, plan 20:00, retain an authoritative organizer answer before final packaging.
Data identity / unit / availability errors	Complete WP02 before stronger observed personalization claims; publish exclusion and reconciliation counts.
Model-family or sensor bias	Develop mixed/omitted mechanisms and shifted sensor relations; publish false-certainty and adverse-case results.
Observation policy does not help	Run isolated failure tests, retain no-acquisition and compare complete policies on a new holdout.
Complex UI obscures meaning	Measure whether reviewers can explain the cutoff, three evidence types, uncertainty and observation rationale.
Submission materials incomplete	Verify every mandatory README item, exact file formats, identity, licensing and signed-out access.
Clinical translation requires partners	Specify collaborator roles, feasibility questions and governed study plans; do not imply that planned studies have happened.

Recommended next sequence

First lock the requirement/claim ledger and the factual endpoint. Then audit data semantics and design the history ablation. In parallel, specify the model graph, observation contracts and isolated delay-value test. Build the unified explanatory experience against those contracts. Freeze new evaluation tasks before tuning. Complete the evidence bundle and full demonstration against the final release.

The next implementation should aim to prove three things: history genuinely changes the model; feasible additional evidence can be evaluated honestly; and the system behaves clearly when available information cannot settle the comparison.

11 / DECISIONS AND NEXT ACTIONS

Terms used throughout the report

Keep the language consistent in code, interface, presentation and evaluation.

Term	Meaning in this dossier
Digital twin	An updating computational patient representation with explicit state, data assimilation and conditional forecasts; the label alone establishes no clinical validity.
Factual forecast	Prediction of a future observed outcome along the actual recorded care path, with inputs restricted to the cutoff.
Hypothetical scenario	A model-produced response under an alternative specified profile. An unchosen outcome is not observed.
Compatibility weight	Relative support within the declared model set, conditional on equations, priors and likelihoods; not a clinical diagnosis probability.
Identifiability	Whether the available observations can distinguish different parameter/state explanations. Good fit does not guarantee this.
Information gain	Expected reduction in uncertainty about the model or state; it can occur without changing a decision.
Decision value	Expected improvement under the declared research objective after information; depends on model assumptions and resources.
Regret	Loss under the selected profile minus the best loss among the declared alternatives under the scoring world.
Calibration / coverage	Agreement of probability/range statements with observed frequencies under the tested setting and assumptions.
Abstention / unresolved	The system withholds a comparison or retains ambiguity because support or evidence is insufficient.
Availability time	When an observation is usable by the algorithm, which can differ from when it was measured.
Clinical benefit	An improvement in clinically relevant outcomes established by an appropriate study; not equivalent to fit, discrimination, information or a successful demo.

Reference register follows. Source notes identify evidence type and boundaries. Local experiment outputs are retained project results, not new studies conducted for this PDF. Proposed work packages and illustrations are authored planning material.

12 / REFERENCE REGISTER

Sources: requirements and clinical context

Primary sources, inspected repository snapshots and retained project evidence.

[R01] Happiest Health - Digital Twin Challenge 2026 Open source
Organizer listing, checked 17 Oct 2026. Source for challenge scope, two-stream requirement, deadlines and deliverables. Numeric judging weights and prior winners of this exact challenge were not verified.

[R02] KDIGO: Blood pressure and volume management in dialysis (2020) Open source
Dialysis-specific conference report. Supports individualized assessment and competing BP/volume objectives. It does not validate RENALIS, its thresholds, observation policy or simulated treatment effects.

[R03] Levin et al.: Hemodynamic response to fluid removal (2018) Open source
Original observational study: 54 patients, 263 sessions; cardiac-power, resistance and combined response patterns. The clinical groups do not validate RENALIS's authored model families or observation equations.

[R04] HEMOBP original data descriptor (2019) Open source
Taiwan outpatient data collected in 2013-2018. Describes 1,075 patients, 165,986 sessions and 4,366,298 machine records. Records are not independent cuff observations. Used for dataset provenance, sampling and source-field interpretation.

[R05] Real-time IDH prediction using ML and cloud infrastructure (2023) Open source
Original historical/device prediction study. Establishes prior work for advance IDH forecasting and two-stream fusion. Its cohort, split, horizon and endpoint differ from our benchmark; reported performance is not directly comparable.

[R06] IoMT and EMR model for hemodialysis complications (2023) Open source
Original integration/prediction paper using human-digital-twin terminology. Relevant priority boundary. It does not establish RENALIS's complete model-disagreement and observation-value workflow.

[R07] BestShape: clinical outcomes after four-year follow-up (2025) Open source
Retrospective evaluation at two Taiwanese institutions. Relevant deployed prediction-and-response comparator. Observational outcome associations are not randomized proof of clinical benefit and do not validate RENALIS.

[R08] B. Braun: Dialog iQ / bioLogic Fusion Open source
Manufacturer description of adaptive UF using systolic BP and relative blood volume. Evidence of product positioning and established functionality; manufacturer statements are not independent comparative effectiveness evidence.

12 / REFERENCE REGISTER

Sources: measurement and evaluation

Primary sources, inspected repository snapshots and retained project evidence.

[R09] Centre-level survival variation in Indian haemodialysis (2024) Open source
Observational private-network cohort of 23,601 patients, excluding the first 90 days at a centre. Supports Indian care relevance, not national prevalence, national IDH incidence or outcome benefit from this project.

[R10] Intradialytic blood-volume monitoring randomized trial (2005) Open source
443-patient trial found no monitoring benefit and observed adverse outcome differences; authors caution about low control-group event rates. Supports separately evaluating information and the policy using it; does not condemn all monitoring.

[R11] Intra-session recurrence and IDH prediction (2026) Open source
Original study dated 7 Jul 2026. Distinguishes initial and recurrent events and demonstrates evaluation sensitivity to recurrence. Motivates separate tasks and a prior-occurrence baseline; findings are dataset-specific.

[R12] Intermittent versus continuous BP monitoring during dialysis (2026) Open source
Original comparison of intermittent cuff and continuous monitoring. Supports explicitly measured endpoints and caution about unobserved intervals. Continuous and cuff measurements are not assumed interchangeable.

[R13] Plasma-volume physiological modeling study (2025) Open source
Original modeling study reporting that improved fit could require physiologically unacceptable parameter values. Motivates plausibility checks and separating fit from mechanism identification.

[R14] Patient-specific relative blood-volume forecasting study (2026) Open source
Original study relevant to patient-wise versus session-wise evaluation and personalization. Does not supply RENALIS's channel-error assumptions or validate its measurement-selection policy.

[R15] Prediction under interventions: counterfactual evaluation (2024) Open source
Original methods for longitudinal counterfactual prediction evaluation. Supports separating factual forecast validation from intervention effects; requires causal assumptions and is not a drop-in short-horizon validation method.

[R16] Optimal perturbation and sampling design for dialysis pharmacokinetics (2016) Open source
Original methods/simulation study; published online in 2015, journal year 2016. Prior art for informative sampling in dialysis-related models, rather than validation of bedside hypotension measurement selection.

12 / REFERENCE REGISTER

Sources: reporting, winners and graph research

Primary sources, inspected repository snapshots and retained project evidence.

[R17] TRIPOD+AI reporting statement (2024) Open source
Primary prediction-model reporting guidance. Useful structure for transparent data, modeling, validation and limitations. Following a reporting checklist is not a claim of validated clinical performance.

[R18] DECIDE-AI guideline (2022) Open source
Primary guidance for early clinical evaluation of AI decision support. Relevant to later appropriately governed clinical workflow studies, not evidence that the current sandbox is suitable for patient care.

[R19] ABDM FHIR implementation guide Open source
Official Indian implementation guide. Planning reference for a small synthetic exchange example. A mapped sample is distinct from validated conformance, certification or live ABDM integration.

[R20] Harvard: HospiTwin award at MIT Hacking Medicine 2025 Open source
Institutional account confirming an InterSystems best-use-of-GenAI challenge award and the ED-boarding use case. It does not establish the overall GrandHack winner or the reasons for jury scores.

[R21] Health in Climate AI: official NYC 2025 winners Open source
Organizer source: CareCast first, Verdant second, HeatIQ third. Also publishes that event's rubric. These results and weights belong to another competition and cannot be transferred as Happiest Health rules.

[R22] Health in Climate AI: official NYC 2026 winners and finalists Open source
Organizer source: Chamomile first, Grounded Cyborg second, HeatSafe third; other finalists include Climate Compass, Five Degrees and CareBuilders. Public descriptions support design comparisons, not causal explanations of judging.

[R23] Barbiero, Vinas Torne and Lio: graph forecasting digital patient (2021) Open source
Peer-reviewed architecture combining graph forecasting, mathematical models and generative integration. Two simulated clinical cases; not clinically validated dialysis forecasting. Primary research inspiration for a structured RENALIS model graph.

[R24] digital-patient: example preprocessing and splitting Open source
Source inspection: repeated simulated trajectories, global scaling before split and shuffled overlapping windows. These example choices do not establish unseen-patient generalization. No code was run or adopted.

12 / REFERENCE REGISTER

Sources: repository audit I

Primary sources, inspected repository snapshots and retained project evidence.

[R25] digital-patient: estimator and Apache-2.0 license Open source
Inspected _patient.py, _nn.py and LICENSE. Uses graph attention; predict contains an example-processing cap. Adaptation needs dependency review, evaluation redesign and retained Apache notices. Package presence alone does not establish calibrated uncertainty.

[R26] LifestyleCorp: medical-digital-twin README Open source
README explicitly describes an incomplete literature-stage repository. Broad whole-body feature descriptions were not treated as implemented functionality. Visual assets require their own provenance and licensing checks.

[R27] LifestyleCorp: simulation service Open source
The inspected file is empty. Selected other backend files contain illustrative functions and generic model code. This source review does not assert that every repository component was executed or tested.

[R28] LifestyleCorp: LICENSE file Open source
Inspected LICENSE is empty despite an MIT statement in the README. Clarify permission before copying code or assets; a public repository and a badge do not resolve all reuse conditions.

[R29] BioTwin AI: digitalTwin.service.js Open source
Inspected risk, progression and treatment-scoring formulas. Includes formula-derived survival/confidence outputs and optional feedback adjustments. This finding concerns that service, not a claim that every application path is mocked.

[R30] ARYANJUMDE GlucoTwin: training and model selection Open source
Actual LR/RF/XGBoost pipeline. Uses stratified random row splitting and picks the best model on the same reported evaluation set. Source-only review; performance was not independently reproduced.

[R31] ARYANJUMDE GlucoTwin: endpoint and scenario implementation Open source
Feature code labels a >=40 mg/dL rise within the future horizon. Twin simulate edits latest-row features and rescores. Such a sensitivity analysis does not itself identify causal intervention effects.

[R32] ARYANJUMDE GlucoTwin: MIT license Open source
Nested GlucoTwin LICENSE contains MIT terms. Retain applicable copyright and permission notices for adopted substantial portions. Verify dependencies and assets separately.

12 / REFERENCE REGISTER

Sources: repository audit II and adjacent work

Primary sources, inspected repository snapshots and retained project evidence.

[R33] Amity GlucoTwin: README and submission artifacts Open source
Includes model artifacts, architecture/presentation files and a demo link. README states synthetic data, random-row evaluation and absent clinical validation. Artifact presence was inspected; video duration and public playback were not reproduced.

[R34] Amity GlucoTwin: exact-horizon target Open source
Code defines glucose >=180 mg/dL exactly eight 15-minute steps later. This differs from a >=40 mg/dL increase anywhere within two hours. Future glucose is a target helper, not an allowed forecast input.

[R35] Amity GlucoTwin: training split Open source
Training excludes patient ID, timestamp, future_glucose_2h and the target, then performs a stratified random row split. Does not establish generalization to unseen patients or future sessions.

[R36] Amity GlucoTwin: MIT license Open source
MIT license inspected. Reuse decisions must retain applicable notices and identify adopted material rather than imply that referenced work is original RENALIS development.

[R37] Anurag ICU twin: prediction integration status Open source
Prediction page explicitly labels model integration pending. Inspected apiService defines a backend contract; monitoring service includes replay and simulated perturbations. Proposed CNN-BiLSTM architecture is not an implemented-model performance result.

[R38] UMCU sasicu: custom license Open source
Custom research/educational terms require prior written permission for reuse and explicitly include AI-system input. Implementation analysis stopped after discovering this. No implementation adoption is included in the plan.

[R39] BioTwin Omni: public project pitch Open source
Self-authored description mentioning this challenge, multi-agent deliberation, simulations and visual features. Treat as positioning evidence; functionality, clinical claims, actual submission and award status are unverified.

[R40] Diaprobe: technology description Open source
Company description of sensing/stimulation for venous-return dynamics; explicitly under development. Adjacent prior-art signal rather than independent performance evidence or permission to perform clinical perturbations.

12 / REFERENCE REGISTER

Sources: further context and local evidence

Primary sources, inspected repository snapshots and retained project evidence.

[R41] Dialysis practices across India: multicentre study Open source
Original practice study documenting variation across participating centres. Motivates studying resource constraints. Our proposed basic/equipped/research configurations are authored scenarios, not measured equipment-prevalence estimates.

[R42] Cardiac-output method validation during dialysis Open source
Method-specific validation study relevant to measurement error. Does not justify universal accuracy assumptions for all noninvasive cardiac-output devices; real-device validation remains necessary.

[R43] Gibbs and Candes: adaptive conformal inference (2021) Open source
Primary methods paper about distribution shift and long-run coverage. A relevant uncertainty-method candidate, not an automatic patient-conditional guarantee or validation of treatment-response scenarios.

[R44] UK Kidney Association dialysis BP guideline (2025) Open source
Dialysis-specific guideline to include in expert review. A clinical reference checkpoint; the report does not prescribe a treatment protocol or claim that RENALIS implements this guideline.

[R45] SICATA haemodialysis survival digital twin abstract (2024) Open source
Conference abstract, weaker evidence than a full clinical evaluation. Establishes earlier use of dialysis digital-twin terminology; task is survival modeling rather than the proposed bedside observation workflow.

[L01] User-supplied organizer brief and submission-form screenshot
Primary supplied requirement documents: attachments/12fe6a42-e1f2-4af4-a7ec-dc049f5a1bde/Pasted text.txt and codex-clipboard-89d83aad-54c0-4170-8670-97da5b608537.png. Screenshot lists all 12 README items as mandatory. Brief and form contain different video-duration language.

[L02] RENALIS observed-baseline.json
Local generated report in docs/observed-baseline.json, inspected 17 Oct 2026. Source of all RENALIS observed-data counts and metrics in this dossier. Checked-in results were inspected, not regenerated for this PDF.

[L03] RENALIS originality-challenges.json
Local generated report in docs/originality-challenges.json: observation-challenges/0.2, 108 cases and 324 policy records. Source of synthetic regret and acquisition counts. No new challenge run was performed for this dossier.

12 / REFERENCE REGISTER

Sources: implementation, conversation and method

Primary sources, inspected repository snapshots and retained project evidence.

[L04] RENALIS model card and research/validation documents
docs/model-card.md; docs/research-review-2026-10.md; docs/execution-and-validation-plan.md; docs/originality.md. Updated 16 Oct 2026. Used to distinguish implemented features, experimental assumptions and future work; historical verification counts are not new test results.

[L05] RENALIS implementation files inspected in the research audit
backend/model.py, backend/mechanisms.py, backend/challenges.py, backend/measurement_fixtures.py and scripts/hemo.py. Referenced to ground current architecture and candidate failure explanations. The delayed-value concern remains a hypothesis requiring an isolated test.

[L06] Conversation and existing video deliverables
Recent project research, repository comparison, pitch guidance and video-clarity requests. Existing v2 film is approximately 2:46 with Kokoro Heart narration. Scope is summarized, not reproduced as a verbatim four-prompt transcript.

Local evidence locations

Project root: D:/Projects/24. ai healthcare project. Primary local paths are relative to this root in the register. Raw permitted data and credentials are not embedded. The user-supplied requirement files are identified by attachment name rather than copied into the public report.

Search and audit boundaries

Focused research covered IDH forecasting, hemodynamic mechanisms, blood-volume monitoring, informative sampling, physiological-model identifiability, counterfactual evaluation, Indian dialysis context, competition results and the seven supplied repositories. Primary papers, official guidance, organizer pages and source files were prioritized. The review is neither exhaustive nor proof of global novelty.

External repository references are pinned to inspected commits where available. UMCU implementation was excluded after license review. Selected competitor code was read only; no external installation, training or clinical result was reproduced. Current RENALIS results were read from retained reports. Product behavior and metrics should be reverified against the eventual submission release.

Report date: 17 October 2026. Classification: internal research strategy and proposed master plan. Figures 1-12 are original explanatory graphics; schematic figures are labeled and do not invent clinical observations. The PDF and its supporting generation/QA files are the only artifacts authored for this request.